MEESAM BILAL

AI / ML Engineer | Computer Vision | NLP, LLMs & Chatbots
Jauharabad, Punjab, Pakistan | +92 302 7344417 | +92 322 9548529
meesambilal786@gmail.com | github.com/MeesamBilal | linkedin.com/in/meesam-bilal-4953b4233

PROFESSIONAL SUMMARY

BS Computer Science (AI specialization) graduate who builds and delivers real AI systems end to end: computer vision, NLP, voice agents and AI web apps. Delivered a paid face-recognition attendance system for a client, completed a hardware-based assistive eyewear final year project, and trained YOLO models reaching 91–97% accuracy on custom datasets. Also builds fast with AI-assisted development (Base44, ChatGPT, Gemini, Claude) to ship working apps and websites. Seeking AI/ML, Computer Vision or Data Science roles and research-oriented opportunities.

EDUCATION

BS Computer Science — Specialization in Artificial Intelligence *2022 – 2026*
PMAS Arid Agriculture University (UIIT), Rawalpindi | CGPA: 3.28 / 4.0

- **Final Year Project:** Multimodal Smart Eyewear, a hardware-integrated assistive device that describes the visual scene aloud (see Projects).
- **Coursework:** Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Reinforcement Learning

PROJECTS

AI-Based Attendance Monitoring System | Paid client project, completed & delivered

- Built a real-time biometric attendance system with Python, OpenCV and FaceNet512, achieving 97%+ face recognition accuracy.
- Implemented one-shot learning so a new user can be enrolled instantly from a single capture, with no retraining.
- Automated attendance logging to CSV, with a thread-safe Tkinter GUI; added voice recognition and TTS for fully touchless operation.

Multimodal Smart Eyewear | Final Year Project, built with hardware

- Designed and built a wearable assistive device combining a computer vision image-to-text pipeline with text-to-speech output.
- Delivers real-time audio descriptions of the surroundings with visual-to-audio latency under 2 seconds, aimed at visually impaired users.
- Integrated software and hardware into a working, demonstrated prototype.

CallBack AI — Missed Call Lead Recovery | Base44, vibe coded

- Designed and built a complete AI web app that helps businesses recover leads they would otherwise lose to missed calls, taking it from idea to a live, working product using prompt-driven (vibe) coding on Base44.
- Planned the app flow, wrote the prompts, tested and refined the result with ChatGPT, Gemini and Claude. Live app

Brew & Bake Concierge | Base44, vibe coded

- Built a complete AI "digital maître d'" for a coffee and bakery business: a chat assistant that handles table reservations, menu and popular-item browsing, coffee, bakery, dessert and meal orders, custom cake requests and opening-hours questions.
- Designed a premium, luxury-style interface and customer experience, developed end to end through prompt-driven (vibe) coding. Live app

Eye Cataract Detection Model | YOLOv11, CVAT

- Collected and annotated 500+ medical fundus images from hospitals using CVAT; trained a YOLOv11 detector with 93% detection accuracy.

Sign Language Detection System | YOLOv8, LabelMe

- Trained a real-time sign detector on a custom dataset of 1,000+ annotated images, reaching 91% mAP on the test set.

Voice Agent System | STT, TTS, LLM

- Built an end-to-end conversational voice agent combining speech-to-text, a GPT-based LLM and text-to-speech for natural multi-turn conversation, with response latency under 1.5 seconds.

Websites & Landing Pages | AI-assisted (vibe coding)

- Designed responsive websites and landing pages, including luxury coffee-shop sites with advanced CSS/JS and Three.js animation, using ChatGPT, Gemini and Claude for rapid development.

Object Dimension Measurement Pipeline | OpenCV, DeepLabV3+, Roboflow

- Built a pipeline with camera calibration, DeepLabV3+ segmentation and pixel-to-mm conversion to measure real object sizes from images; dataset labeled in Roboflow.

Text Summarization & Sentiment Analysis Pipeline | Hugging Face, BERT, T5

- Built an NLP pipeline for classification, summarization and sentiment analysis over 100+ documents with 88% sentiment accuracy.

WORK EXPERIENCE

AI Engineer Intern *Jun 2024 – Aug 2024*

Devrolin Labs — Rawalpindi, Pakistan

- Built end-to-end voice agent pipelines integrating STT, TTS and LLMs, cutting manual interaction time by ~60% in demo environments.
- Developed multi-turn conversational AI with context retention across 10+ dialogue turns using LangChain and GPT-based models.
- Integrated speech processing, NLU and response generation into one voice solution, delivered within a 3-month sprint.

Machine Learning Intern *2023*

Rhombix Technologies — Remote

- Cleaned and preprocessed datasets of 10,000+ samples for training and fine-tuning across several ML projects.
- Implemented and benchmarked 5+ algorithms (classification, regression, clustering), improving baseline accuracy by up to 15%.

Web Development (Frontend Intern & WordPress Developer) *Dec 2022 – 2023*

Youngdev (Remote) and Vitech Software Company (Jauharabad)

- Built 4+ responsive landing pages and a reusable UI component library; launched 2 WooCommerce stores (khaari.pk, bagtronic.com).

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, HTML, CSS
- **ML / DL:** PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers (BERT, T5), LangChain, RAG pipelines, LLM fine-tuning
- **Computer Vision:** YOLOv8, YOLOv11, OpenCV, MediaPipe, FaceNet512, Object Detection, Segmentation, OCR
- **Speech & NLP:** Speech-to-Text, Text-to-Speech, Voice Agents, Sentiment Analysis, Summarization, Machine Translation
- **Deployment & Tools:** FastAPI, Docker, REST APIs, real-time inference, Git/GitHub
- **AI-Assisted Development:** Base44, ChatGPT, Gemini, Claude
- **Data & Annotation:** Pandas, NumPy, CVAT, Roboflow, LabelMe
- **Productivity:** MS Office (Word, Excel, PowerPoint)